

AV NETWORK DESIGN REPORT

Example Hotel Ballroom AV

September 30, 2026

Americas · Catalog 2026.09.27 · Request R-EXAMPLE

VALIDATED BY THE PRO AV DESIGN TEAM

Reviewed by Pro AV Design team, September 30, 2026

82

DEVICES

8

SWITCHES

1,326 W

POE LOAD

2.6 kW

EST. POWER

11.0 U

RACK SPACE

Design summary

This design was validated by the Pro AV Design team on September 30, 2026. Pricing and availability are provided by NETGEAR or your distributor.

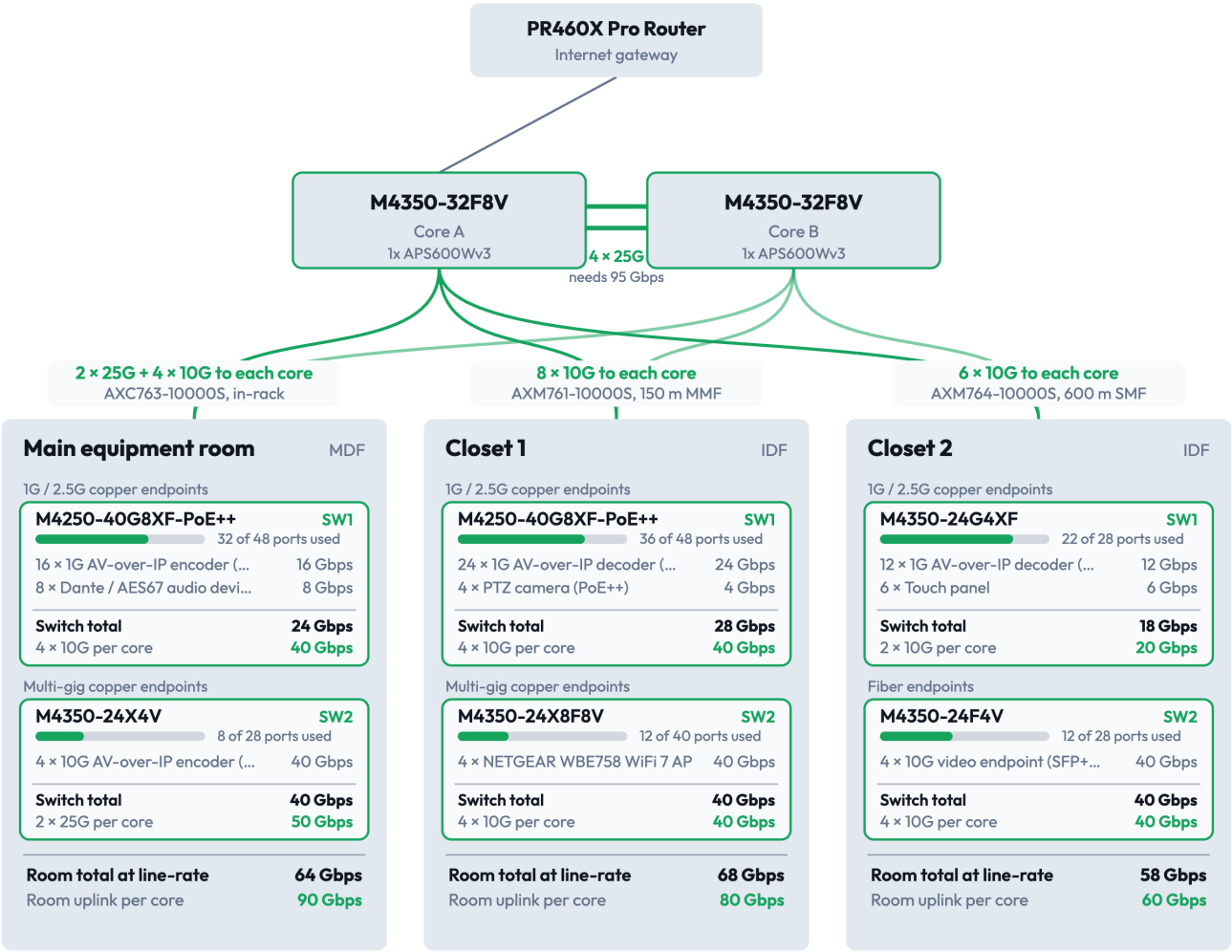
DESIGN SETTINGS

Switch series	Best fit	Spare ports	10%
Core	2 × M4350-32F8V (4 × 25G between cores)	PoE headroom	20%
Redundant core	Yes	Mains voltage	110 V
Redundant power	All switches	Gateway	PR460X Pro Router
Design basis	Line-rate (non-blocking)	OnCall 24x7 support	Not included
Oversubscription	1:1	TAA required	No

ROOMS AND DEVICES

Device	Category	Qty	Link	PoE each
Main equipment room • MDF • 28 devices				
1G AV-over-IP encoder (TX)	Video	16	1G Copper	13 W
10G AV-over-IP encoder (copper)	Video	4	10G Copper	–
Dante / AES67 audio device	Audio	8	1G Copper	13 W
Closet 1 • IDF, 150 m multimode fiber • 32 devices				
1G AV-over-IP decoder (RX)	Video	24	1G Copper	13 W
PTZ camera (PoE++)	Camera	4	1G Copper	60 W
NETGEAR WBE758 WiFi 7 AP	Wireless	4	10G Copper	39 W
Closet 2 • IDF, 600 m single mode fiber • 22 devices				
1G AV-over-IP decoder (RX)	Video	12	1G Copper	13 W
10G video endpoint (SFP+ fiber)	Video	4	10G Fiber	–
Touch panel	Control	6	1G Copper	25 W

Network diagram



Green lines are uplinks, labelled with link count, speed and the optic or cable used. With a redundant core, every switch has its own full-capacity link group to each core, so either core can carry all traffic alone.

Bill of materials

Part number	Description	Qty	Where
SWITCHES			
GSM4248UX-100NAS	M4250-40G8XF-PoE++ switch	2	Main equipment room, Closet 1
GSM4328-100NES	M4350-24G4XF switch	1	Closet 2
XSM4328CV-100NES	M4350-24X4V switch	1	Main equipment room
XSM4328FV-100NES	M4350-24F4V switch	1	Closet 2
XSM4340FV-100NES	M4350-32F8V core switch	2	Main equipment room (core)
XSM4340V-100NES	M4350-24X8F8V switch	1	Closet 1
POWER			
APS350W-100NES	350W modular PSU	2	Main equipment room, Closet 2
APS600W-200NES	600W modular PSU (v2)	1	Closet 2
APS600W-300NES	600W modular PSU (v3)	3	Closet 1, Main equipment room (core)
OPTICS			
AXM761-10000S	10GBASE-SR SFP+, multimode	36	Closet 1, Closet 2
AXM764-10000S	10GBASE-LR LITE SFP+, single mode	24	Closet 2
CABLING			
AXC763-10000S	10G SFP+ DAC 3m, passive copper	9	Main equipment room
To be confirmed	25G direct-attach cable (not in catalog)	8	Main equipment room, Main equipment room (core)
WIRELESS			
WBE758-111NAS	NETGEAR WBE758 WiFi 7 AP	4	Closet 1
GATEWAY			
PR460X-111NAS	PR460X Pro Router	1	Main equipment room

Part numbers are for Americas. Pricing and availability are provided by NETGEAR or your distributor after validation.

Power and rack space

Room	Est. draw	Datasheet max	Heat load	Current	Circuits	UPS size	Rack
Main equipment room	705 W	4,587 W	2,404 BTU/hr	6.4 A at 110 V	1 × 20 A	783 VA+	4.0 U
Closet 1	1,021 W	5,580 W	3,484 BTU/hr	9.3 A at 110 V	1 × 20 A	1,135 VA+	3.0 U
Closet 2	539 W	991 W	1,840 BTU/hr	4.9 A at 110 V	1 × 20 A	599 VA+	2.0 U
Main equipment room (core)	313 W	313 W	1,067 BTU/hr	2.8 A at 110 V	1 × 20 A	348 VA+	2.0 U

POWER SUPPLIES PER SWITCH

Room	Device	Qty	PoE load each	Power supplies	PoE available (1 PSU failed)	Headroom	Est. draw each
Main equipment room	M4250-40G8XF-PoE++	1	312 W	2 of 3 PSUs powered	720 W	57%	429 W
Main equipment room	M4350-24X4V	1	–	1x APS350W	700 W	100%	119 W
Closet 1	M4250-40G8XF-PoE++	1	552 W	2 of 3 PSUs powered	720 W	23%	696 W
Closet 1	M4350-24X8F8V	1	156 W	1x APS600Wv3	290 W	46%	325 W
Closet 2	M4350-24G4XF	1	306 W	1x APS600Wv2	468 W	35%	420 W
Closet 2	M4350-24F4V	1	–	1x APS350W	–	–	119 W
Main equipment room (core)	M4350-32F8V	2	–	1x APS600Wv3	–	–	156 W
Main equipment room	PR460X Pro Router	1	–	Internal PSU	–	–	156 W

Estimated draw = datasheet maximum without PoE + PoE load ÷ 0.9 (PSU efficiency). Circuits assume 80% continuous load. Feed each power supply from a separate circuit or UPS.

Checks

CHECK	Main equipment room: no 25G cable in the catalog for 3 m. Placeholder added.
NOTE	Inter-core link sized for half of all room traffic (active/active cores), and never below the busiest single switch: needs 95 Gbps, provided $4 \times 25G = 100$ Gbps.
CHECK	No 25G inter-core cable in the catalog. Placeholder added.
NOTE	Devices above 30W need Ultra90 PoE++ (802.3bt) ports; only those switches were considered for them.
NOTE	Switches interconnect with LAG uplinks and IGMP Plus (no stacking), which keeps AVB and PTP available.
NOTE	Uplink LAGs use 1, 2, 4 or 8 links to each core, and each core link group carries the switch's full load on its own so one core can fail (sizes from Tool_Settings: 1, 2, 4, 8), so LAG hashing spreads traffic evenly.
NOTE	Redundant power: PoE is sized to the budget that remains if one power supply fails.

UPLINK PLAN

Room	Switch	Units	Uplinks each	Speed	Run	Optic / cable
Main equipment room	M4250-40G8XF-PoE++	1	8	10G	in-rack	AXC763-10000S
Main equipment room	M4350-24X4V	1	4	25G	in-rack	To be confirmed
Closet 1	M4250-40G8XF-PoE++	1	8	10G	MMF, 150 m	AXM761-10000S
Closet 1	M4350-24X8F8V	1	8	10G	MMF, 150 m	AXM761-10000S
Closet 2	M4350-24G4XF	1	4	10G	SMF, 600 m	AXM764-10000S
Closet 2	M4350-24F4V	1	8	10G	SMF, 600 m	AXM764-10000S